

Patient Name : BANDARU RAMESH

Age / Gender : 56 years / Male

Patient ID : 687

Source : DIRECT

Referral : Dr. V.MADHAVA RAO

Collection Time : Sep 15, 2023, 06:20 a.m.

Receiving Time : Sep 15, 2023, 06:28 a.m.

Reporting Time : Sep 15, 2023, 09:06 a.m.

Sample ID :



0232580010

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
PSA			
*Prostate Specific Antigen	0.89	0.0 - 4.0	ng/mL
Method:Chemiluminiscence/Serum			
Instrument	CLIA		
Note	PSA Ratio (Free PSA/Total PSA) is advisable for patient having borderline PSA levels		
Note			

C-Peptide Serum Fasting

*C-Peptide (CPS)	1.01	0.81-3.85	ng/mL
Method : Chemiluminiscence			

NOTE

In normal individuals, the molar ratio of C-peptide to insulin in the fasting state is 5:1

CLINICAL UTILITY

- Assess pancreatic islet cell function
- Distinguish insulin secreting tumors (Insulinoma) from exogenous insulin administration as a cause of hypoglycemia (commercial insulin does not contain C-peptide). Sera from Insulinoma patients have high insulin and high C-peptide levels whereas hypoglycemia from injected or exogenous insulin shows high insulin and low C-peptide levels
- Distinguish Type I and Type II Diabetes mellitus

INCREASED LEVELS:	Insulinoma & Type II Diabetes
DECREASED LEVELS:	Type I Diabetes & Exogenous insulin administration

VITAMIN B12

Vitamin B12	108	222-1439	pg/mL
CLIA			
Method : CLIA ChemiLuminescence Immunoassay			

Method: Chemi Luminescence Immunoassay

Thyroid Function Test

*T3	113.44	87 - 178	ng/dL
*T4	9.18	4.82 - 15.65	µg/dL
*TSH	1.918	0.38 - 5.33	µIU/mL

Method: Chemi Luminescence Immunoassay(Sample Type:Serum)

Interpretation

- Measurement of TSH in conjunction with T3,T4 or FT3,FT4 is used for assessment of thyroid gland functional status and also for monitoring patients on treatment for hyperthyroidism & hypothyroidism. Suppressed TSH (<0.01 µIU/mL) suggests a diagnosis of hyperthyroidism and elevated concentration (>7 µIU/mL) suggest hypothyroidism. TSH levels may be affected by acute illness and several medications including

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dopamine and glucocorticoids.TSH concentration are decreased (low or undetectable) in Graves disease. Increased in TSH secreting pituitary adenoma (secondary hyperthyroidism); PPTH. TSH concentration are primary elevated in hypothyroidism (along with decreased T4) except for pituitary & hypothalamic disease.			
- Mild to modest TSH elevations in patient with normal T3 & T4 levels indicates impaired thyroid hormone reserves & incipient hypothyroidism (subclinical hypothyroidism). - Mild to modest decrease in TSH with normal T3 & T4 indicates subclinical hyperthyroidism. - Degree of TSH suppression does not reflect the severity of hyperthyroidism, therefore, measurement of free thyroid hormone levels is required in patient with a suppressed TSH level. - If free T4 is normal, free T3 should be checked as it is the first hormone to increase in early hyperthyroidism.Though TSH levels can also be used to effectively monitor patients being treated with thyroid hormones but it may be misleading. Therefore total or free T4 generally serve as front line assay during this period. The free T3 & T4 (FT3 & FT4) measures concentrations of free hormones, which are not affected by changing in concentrations of binding proteins, therefore more reliable indicator of true thyroid status.			

VITAMIN D

Vitamin D	25.47	Deficiency <20 Insufficiency 20–30 Sufficiency 30–100 Toxicity > 100	ng/mL
Method:CLIA(Sample Type:Serum)			

Interpretation :

- Vitamin D is a fat soluble vitamin and exists in two main forms as cholecalciferol(vitamin D3) which is synthesized in skin from 7-dehydrocholesterol in response to sunlight exposure & Ergocalciferol(vitamin D2) present mainly in dietary sources.Both cholecalciferol & Ergocalciferol are converted to 25(OH)vitamin D in liver.
- Testing for 25(OH)vitamin D is recommended as it is the best indicator of vitamin D nutritional status as obtained from sunlight exposure & dietary intake. For diagnosis of vitamin D deficiency it is recommended to have clinical correlation with serum 25(OH)vitamin D, serum calcium, serum PTH & serum alkaline phosphatase.
- During monitoring of oral vitamin D therapy- suggested testing of serum 25(OH)vitamin D is after 12 weeks or 3 mths of treatment. However, the required dosage of vitamin D supplements & time to achieve sufficient vitamin D levels show significant seasonal(especially winter) & individual variability depending on age, body fat, sun exposure, physical activity ,genetic factors(especially variable vitamin D receptor responses), associated liver or renal disease, malabsorption syndromes and calcium or magnesium deficiency influencing the vitamin D metabolism Vitamin D toxicity is known but very rare.kindly correlate clinically, repeat with fresh sample if indicated.

Associated Test Profile :

- For diagnosis of vitamin D deficiency it is recommended to have clinical correlation with serum 25(OH)vitamin D and serum PTH.An inverse relationship exists between PTH and 25(OH)D levels, Parathyroid hormone levels start to rise at 25(OH)D levels below 31 ng/mL & usually decrease after the correction of vitamin D insufficiency.Thus, restoration of PTH and 25(OH) D levels to normalcy after adequate vitamin D replacement therapy is a useful monitoring strategy.
- As a holistic & scientific approach for diagnosis and optimal treatment for vitamin D deficiency, Vitamin D plus profile (25 Hydroxy(OH) Vit D and PTH) is suggested.

Note

****END OF REPORT****

Approved By

Dr. V. Madhava Rao

MD (PGI-Chandigarh)

DM (Endocrinology) (AIIMS-New Delhi)

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)
<u>HEMOGRAM (CBP)</u>			
*Haemoglobin	13.5	12.0 - 18.0	g/dL
*R.B.C count	4.92	3.80 - 5.30	10 ⁶ μL
*Leucocyte Count (WBC Count)	7300	4000 - 11000	μL
*Platelet Count	198000	120000 - 380000	μL
*PCV	41.5	36.0 - 56.0	%
*MCV	84.3	80.0 - 100.0	fL
*MCH	27.4	27-32	pg
*MCHC	32.5	32.0 - 36.0	g/dL
*RDW-CV	12.7	10.0 - 16.5	%
*Mean Platelet Volume (MPV)	7.7	5.0 - 10.0	fL
*PDW	16.5	12.0 - 18.0	%
<u>*DIFFERENTIAL COUNT</u>			
*Granulocyte	75.2	42 - 85	%
*Lymphocytes	21.0	11.0 - 49.0	%
*Monocytes	3.8	0.0 - 9.0	%

Instrument:TEST PERFORMED ON NIHON KOHDEN HEMATOLOGY ANALYZER

Note

S.PHOSPHORUS

PHOSPHORUS	2.95	2.5 - 4.5	mg/dL
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Method: PHOSPHOMOLYBDATE/UV (Sample Type:Serum)

Method:TEST DONE ON BIOSYSTEMS BA 400. A FULLY AUTOMATED BIOCHEMISTRY ANALYSER

E S R

*ESR	12	0 - 15	mm/Hour
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Method Automated Westergren

Interpretation:

- It indicates presence and intensity of an inflammatory process. It does not diagnose a specific disease. Changes in the ESR are more significant than the abnormal results of a single test.
- It is a prognostic test and used to monitor the course or response to treatment of diseases like tuberculosis, bacterial endocarditis, acute rheumatic fever, rheumatoid arthritis, SLE, Hodgkins disease, temporal arteritis and polymyalgia rheumatica.
- It is also increased in pregnancy, multiple myeloma, menstruation, and hypothyroidism.

Note

URINE ALBUMIN/CREATININE RATIO

*URINE ALBUMIN	22.95	0 - 15	mg/L
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Latex Method(Sample Type:Urine)

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)
*URINE CREATININE Jaffe's Compensated(Sample Type:Urine)	106.13	37 - 250	mg/dL
*URINE ALBUMIN/CREATININE RATIO calculated	21.62	0 - 30	mg/gram creatinine

Instrument: Tests performed on Biosystems BA 400, A Fully Automatic Biochemistry analyser

Normal: 0 - 30

Microalbuminuria: 30 - 300 mg/gram creatinine

Macroalbuminuria : > 300 mg/gram creatinine

Note

COMPLETE URINE EXAMINATION

Colour (Sample Type:Urine)	Yellow	Pale yellow
pH	5.5	4.5 – 8.5
Specific Gravity	1.020	1.015 – 1.025
Glucose	Trace	Negative
Protein	Negative	Negative
Ketones	Negative	Negative
Bilirubins	Negative	Negative
Nitrite	Negative	Negative
Blood	Negative	Negative
Urobilinogen	Negative	Negative
Leukocytes	Negative	Negative

MICROSCOPIC EXAMINATION

Pus cells (WBCs)	2 - 3	0 - 2/hpf	/hpf
Epithelial cells	1 - 2	0 - 4/hpf	/hpf
Red blood cells	Nil	0 - 2/hpf	/hpf
Cast	Nil	Nil	
Crystals	Nil	Nil	
Other	Nil	Nil	

Note

FASTING PLASMA GLUCOSE

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)
Fasting Plasma Glucose (F) Method : Glucose oxidase/Peroxidase	166.07	Desired Fasting Plasma Glucose: For nondiabetic: 70 - 100 mg/dL For Diabetic: 80 - 130 mg/dL	mg/dL
<u>Liver Function Tests</u>			
BILIRUBIN (TOTAL) DICHLOROPHENYL DIAZONIUM (Sample Type:Serum)	0.54	<24Hrs: 1.0 - 8.0 mg/dL <48Hrs: 6.0 - 12.0 mg/dL 3 - 5 Days: 10.0 - 14.0 mg/dL Adults : <1.0 mg/dL	mg/dL
*BILIRUBIN (DIRECT) DICHLOROPHENYL DIAZONIUM (Sample Type:Serum)	0.24	< 0.2 mg/dL	mg/dL
BILIRUBIN (INDIRECT)(Sample Type:Serum) calculated Method : Calculated	0.30	0.2 -0.8 mg/dL	mg/dL
SGOT IFCC without pyridoxal 5-P(Sample Type:Serum)	13.32	0 - 40	U/L
SGPT IFCC without pyridoxal 5-P(Sample Type:Serum)	16.88	0 - 41	U/L
TOTAL PROTEIN BIURET (Sample Type:Serum)	7.72	Pre mature : 0.6 -6.0 g/dL New born : 4.6 - 7.0 g/dL One week : 4.4 - 7.6 g/dL 7 months - 1 year : 5.1-7.3 g/dL 1 Year - 2 years : 5.6 - 7.5 g/dL > 3 years : 6.5 - 8.0 g/dL	g/dL
ALBUMIN BROMOCRESOL GREEN(Sample Type:Serum)	4.55	3.5 - 5.0 g/dL	g/dL
*GLOBULIN Calculated	3.17	2.5 - 4.5	g/dL
*A:G RATIO	1.44	1.5 - 3.5	
ALP (ALKALINE PHOSPHATASE) 2Amino 2Methyl - 1Propanol Buffer IFCC Method(Sample Type:Seru	79.52	<115 U/L	U/L
*gGT Method : IFCC	18.68	< 55	U/L
<u>LIPID PROFILE</u>			
TOTAL CHOLESTEROL Method:Cholestrol Oxidase/Peroxidase(Sample Type:Serum)	130.38	Desirable : Up to 200 mg/dL Borderline High: 200-239 mg/dL High : > 240 mg/dL	mg/dL

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TRIGLYCERIDES Method:Glycerol phosphate oxidase/Peroxidase(Sample Type:Serum)	90.10	Normal : Up to 150 mg/dL Borderline High : 150-199 mg/dL High : 200-499 mg/dL Very high : > 500 mg/dL	mg/dL
HDL CHOLESTEROL Method:Direct(Sample Type:Serum)	41.31	> 40 mg/dL High Risk : < 35 mg/dL Low Risk : > 60 mg/dL	mg/dL
*LDL CHOLESTEROL Calculated	71.05	Optimal : Up to 100 mg/dL Near optimal/above optimal : 100-129 mg/dL Borderline High : 130-159 mg/dL High : 160-189 mg/dL Very High : > 190 mg/dL	mg/dL
*VLDL CHOLESTEROL Calculated	18.02	5 - 51 mg/dL	mg/dL
*CHOL / HDL RATIO Calculated	3.16	Low risk : < 3.5 Average risk : 3.5 - 5.0 High risk : > 5.0	
*LDL/HDL RATIO Calculated	1.72	Desirable : < 0.5-3.0 Borderline : 3.1- 6.0 Elevated : > 6.0	

INSTRUMENT: DONE ON BA 400 BIOSYSTEMS FULLY AUTOMATED BIOCHEMISTRY ANALYZER

Note

S.IRON

IRON	62.98	Men : 65 - 175 µg/dL Women : 50 - 170 µg/dL	µg/dL
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Method:FERROZINE(Sample Type:Serum)

Instrument:Test Done On Fully Automated Biochemistry Analyser, BIOSYSTEMS BA 400

Note

URIC ACID

URIC ACID	4.52	Male:3.5-7.2 Female:2.6 - 6.0	mg/dL
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Method:URICASE/PEROXIDASE(Sample Type:Serum)

Note

UREA

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)
UREA Method:UREASE / GLUTAMATE DEHYDROGENASE(Sample Type:Serum)	28.37	Cord: 21 - 40 mg/dL Pre - Mature : 3 - 20 mg/dL New Born : 4 - 12 mg/dL Child : 5 - 18 mg/dL Adult : 15 - 39 mg/dL <60 Years : 8 - 23 mg/dL	mg/dL

Note

S.CREATININE

CREATININE Method:JAFTE COMPENSATED (Sample Type:Serum)	0.76	1 - 4 Day's : 0.3 - 1.0 mg/dL Infants : 0.2 - 0.4 mg/dL Child : 0.3 - 0.7 mg/dL Adults : (M) : 0.7 - 1.2 mg/dL Adults : (F) : 0.5 - 0.9 mg/dL	
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Note

HbA1c

HbA1c Method:HPLC (BIORAD D 10)(Sample Type:Blood)	8.6	Non Diabetic : 4.0 - 5.6 % Border line : 5.7 - 6.4 % Good control in Diabetes : < 6 Years : < 8.5 % 6 Years - 12 Years : < 8.0 % 13 Years - 18 Years : < 7.5 % > 18 Years : < 7 %	%
Estimated Average Glucose (eAG) Calculated	200.12		mg/dL

Interpretation :

- HbA1c is used for monitoring diabetic control. It reflects the estimated average glucose (eAG).
- HbA1c has been endorsed by clinical groups & ADA (American Diabetes Association) guidelines 2012, for diagnosis of diabetes using a cut-off point of 6.5%. ADA defined biological reference range for HbA1c is 4% to 5.7%. Patients with HbA1c value between 5.7% to 6.5% are considered Pre-diabetic.
- Trends in HbA1c are a better indicator of diabetic control than a solitary test.
- Low glycated haemoglobin(below 4%) in a non-diabetic individual are often associated with systemic inflammatory diseases,chronic anaemia(especially severe iron deficiency &haemolytic), chronic renal failure and liver diseases. Clinical correlation suggested.
- To estimate the eAG from the HbA1C value, the following equation is used: $eAG(mg/dl) = 28.7 \times A1c - 46.7$
- Interference of Haemoglobinopathies in HbA1c estimation.
 - For HbF > 25%, an alternate platform (Fructosamine) is recommended for testing of HbA1c.
 - Homozygous hemoglobinopathy is detected, fructosamine is recommended for monitoring diabetic status

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C. Heterozygous state detected is corrected for HbS and HbC trait.

Hemoglobin electrophoresis (HPLC method) is recommended for detecting hemoglobinopathy.

Note

POST PRANDIAL PLASMA GLUCOSE

Post Prandial Plasma Glucose	238.20	< 180	mg/dL
Method : Glucose oxidase/Peroxidase			

Instrument: Done on BIOSYSTEMS BA 400, a Fully Automated Biochemical Analyser

Note

Electrolytes

Sodium	139.8	136 - 145	mmol/L
Method: ISE Direct(Sample Type: Serum)			
Potassium	4.46	3.3 - 5.1	mmol/L
Method: ISE Direct(Sample Type: Serum)			
Ionised calcium	4.86	4.5 - 5.3	mg/dL
Method: ISE Direct(Sample Type: Serum)			
Chloride	99.2	90 - 110	mmol/L
Method: ISE Direct(Sample Type: Serum)			

Instrument: Test Done On Fully Automated Biochemistry Analyser, Sensa Core ST-200

Note

END OF REPORT

Approved By

Dr. V. Madhava Rao

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A. Kiran Kumar
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END OF REPORT